

## Part C – Ethical considerations when designing AI systems

- **Due** 11:59pm Sunday 1 June 2025 (Week 12)
- **Contributes** 10% to your Assignment 2 result.
- **Individual**.

*Part C of Assignment 2 (2C) requires you to work individually to explore ethical considerations when making technical and design decisions while developing the TBRGS.*

In Part C of Assignment 2, you will work individually. During your participation and involvement in the development of the TBRGS you have encountered a number of design decisions and technical choices that can potentially have some ethical impacts on the product and its users. Your task in Part C (2C) is to identify such design and technical decisions and their potential ethical implications. By documenting and performing an analysis into these decisions and their ethical implications, you can help the development team have a deeper understanding into these issues and see the alternatives to make informed decisions on the design and implementation of the product.

For this document, you will consider the TBRGS system developed by your team and a potential TBRGS+ system that can be developed for the intended functions. For instance, TBRGS+ can be an app run on mobile devices (similar to Google Maps) that can collect personal information such as their locations and even their user profiles (including their personal preferences).

### Objectives of the Assignment

On successful completion of Part C, you will be able to demonstrate your ability to apply ethical considerations when designing and developing an AI system. More specifically, your report will:

1. *Identify the key issues related to an AI system you and your team are developing as well as the advanced product that you may design if you are provided with sufficient time and resources*; that is, you are able to identify and describe the design decisions and their potential ethical implications for this product and its possible extensions.
2. *Demonstrate the ability to analyse relevant aspects of these decisions as well as their pros and cons*; that is, you are able to describe the alternatives for each decisions and the impacts they would have on the users and the product's owner and developer.
3. *Provide clear conclusions and recommendations*; that is, you can present your analysis and arguments in a convincing manner to ensure that clear conclusions and perhaps recommendation can be made.

### Submission instructions

Your report has to be either in Microsoft Word or in PDF whose name is your Student ID (for example, 1234567.PDF). You must submit your work via Canvas.

Standard late penalties apply - 10% for each day late, more than 5 days late is 0%.  
You can follow the following structure for the final report:

- **Cover page:** including your student details (i.e., your full name and student ID).
- **Table of contents (TOC).**
- **Executive Summary:** (2 paragraphs) First paragraph provides a summary of the potential ethical issues associated with TBRGS or TBRGS+. Second paragraph summarises your analyses and conclusions/recommendations.
- **Main report:** (1,500-2,000 words)
  - **Introduction:** Provide a discussion on the design decisions for TBRGS and TBRGS+ and the potential ethical issues associated with these decisions.
  - **Findings:** Describe the alternatives for each decisions and the impacts they would have on the users and the product's owner and developer.
  - **Conclusion and recommendations:** Give a summary of the analyses presented earlier in your report to allow you to arrive at the conclusions and recommendations you want to make.

- **Acknowledgements/Resources:** Include in your report a list of the resources you have used to create your work. A simple list of URL's is not enough. Include with each entry a basic description of how the person/AI tool or website assisted you in your work.
- **References:** Cite the sources you referred to in your report.

**Tips:**

- If you choose to include a picture/diagram, one figure is equivalent to 50 words.
- Report presentation should include header/footer information (pages numbers, etc.)

## Marking Scheme

| <b>Requirements</b> (We adapt the <b>Research Paper Rubric</b> from <b>Cornell College</b> for a more specific <b>rubric</b> for this report. See next page.) | <b>Mark</b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Content</b>                                                                                                                                                | <b>75</b>   |
| <b>Presentation</b>                                                                                                                                           | <b>25</b>   |
| <b>Total</b>                                                                                                                                                  | <b>100</b>  |

**Adapted from Cornell College's Research Paper Rubric**

(<https://www.cornellcollege.edu/library/faculty/focusing-on-assignments/tools-for-assessment/ResearchPaperRubric.pdf>)

|                     | EXPERT                                                                                                                                                                                                                                                                                                                      | PROFICIENT                                                                                                                                                                                                                                                                                                             | APPRENTICE                                                                                                                                                                                                                                         | NOVICE                                                                                                                                                                                                        |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TOPIC COVERAGE      | The report covers at least two design decisions connecting to both Part A and Part B (for TBRGS) and also at least two additional design decisions related to TBRGS+. The report demonstrates comprehensive understanding of these decisions (including the alternatives and their ethical implications among other things) | The report covers at least two design decisions connecting to both Part A and Part B (for TBRGS) and also at least one additional design decisions related to TBRGS+. The report demonstrates a decent understanding of these decisions (including the alternatives and their ethical implications among other things) | The report covers at least two design decisions connecting to both Part A and Part B (for TBRGS). The report demonstrates a decent understanding of these decisions (including the alternatives and their ethical implications among other things) | The paper does not demonstrate that the author has fully understood and identified appropriate design decisions and/or their alternatives and/or the ethical implications of different alternatives .         |
| DEPTH OF DISCUSSION | In-depth discussion & elaboration in all sections of the paper.                                                                                                                                                                                                                                                             | In-depth discussion & elaboration in most sections of the paper.                                                                                                                                                                                                                                                       | The writer has omitted pertinent content or content runs-on excessively. Quotations from others outweigh the writer's own ideas excessively.                                                                                                       | Cursory discussion in all the sections of the paper or brief discussion in only a few sections.                                                                                                               |
| COHESIVENESS        | Ties together information from all sources. Paper flows from one issue to the next without the need for headings. Author's writing demonstrates an understanding of the relationship among material obtained from all sources.                                                                                              | For the most part, ties together information from all sources. Paper flows with only some disjointedness. Author's writing demonstrates an understanding of the relationship among material obtained from all sources.                                                                                                 | Sometimes ties together information from all sources. Paper does not flow - disjointedness is apparent. Author's writing does not demonstrate an understanding of the relationship among material obtained from all sources.                       | Does not tie together information. Paper does not flow and appears to be created from disparate issues. Headings are necessary to link concepts. Writing does not demonstrate understanding any relationships |
| SPELLING & GRAMMAR  | No spelling &/or grammar mistakes.                                                                                                                                                                                                                                                                                          | Minimal spelling &/or grammar mistakes.                                                                                                                                                                                                                                                                                | Noticeable spelling & grammar mistakes.                                                                                                                                                                                                            | Unacceptable number of spelling and/or grammar mistakes.                                                                                                                                                      |
| SOURCES             | More than 5 current sources, of which at least 3 are peer-review journal articles or scholarly books. Sources include both general background sources and specialized sources. Special-interest sources and popular literature are acknowledged as such if they are cited. All web sites utilized are authoritative.        | 5 current sources, of which at least 2 are peer-review journal articles or scholarly books. All web sites utilized are authoritative.                                                                                                                                                                                  | Fewer than 5 current sources, or fewer than 2 of 5 are peer-reviewed journal articles or scholarly books. All web sites utilized are credible.                                                                                                     | Fewer than 5 current sources, or fewer than 2 of 5 are peer-reviewed journal articles or scholarly books. Not all web sites utilized are credible, and/or sources are not current.                            |
| CITATIONS           | Cites all data obtained from other sources. APA citation style is used in both text and bibliography.                                                                                                                                                                                                                       | Cites most data obtained from other sources. APA citation style is used in both text and bibliography.                                                                                                                                                                                                                 | Cites some data obtained from other sources. Citation style is either inconsistent or incorrect.                                                                                                                                                   | Does not cite sources.                                                                                                                                                                                        |

**Content** = TOPIC COVERAGE + DEPTH OF DISCUSSION + COHESIVENESS

**Presentation** = SPELLING & GRAMMAR + SOURCES + CITATIONS